Name: Rahul K Roll: DA24B042

Q1. c

Q2. a

Q3. out = (W - K)/S + 1

= (64 - 5)/2 + 1 = 29.5 + 1 = 30

shape = 30 x 30 x 16

params = 5x5x3x16 + 16 = 1216

Q5. $z = wx + b$, $y = \text{sigmoid}(z)$

~~$dL/dw = (y - t)\,x$~~

$dL/dw = dL/dy \cdot dy/dz \cdot dz/dw$

$dL/dy = (y - t)$

$dy/dz = \text{sigmoid}'(z)$

so $dL/dw = (y - t)\ \text{sigmoid}'(z)\ x$

Q4. flatten = 30 x 30 x 16 = 14400

weights = 14400 x 10 = 144000

+ 10 bias = 144010

Q7. def softmax(x):

```
    import math
    exps = [math.exp(v) for v in x]
    total = sum(exps)
    return [e/total for e in exps]
```

Q6. Dropout randomly turns off neurons during training with probability p. This stops overfitting because the network can not depend on one neuron.